BABAK ESMAEILI

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PUBLICATIONS

Conference

[1] H. Zimmermann, H. Wu, B. Esmaeili, and J.-W. van de Meent. "Nested Variational Inference". In: *Advances in Neural Information Processing Systems*. Ed. by A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan. 2021.

[2] H. Wu*, B. Esmaeili*, M. Wick, J.-B. Tristan, and J.-W. Van De Meent. "Conjugate Energy-Based Models". In: *Proceedings of the 38th International Conference on Machine Learning*. Ed. by M. Meila and T. Zhang. Vol. 139. Proceedings of Machine Learning Research. PMLR, 2021, pp. 11228–11239.

[3] A. Bozkurt*, B. Esmaeili*, D. H. Brooks, J. G. Dy, and J.-W. van de Meent. "Rate-Regularization and Generalization in VAEs". In: *The 24th International Conference on Artificial Intelligence and Statistics*. 2021.

[4] B. Esmaeili, H. Wu, S. Jain, A. Bozkurt, N. Siddharth, B. Paige, D. H. Brooks, J. Dy, and J.-W. Meent. "Structured Disentangled Representations". In: *The 22nd International Conference on Artificial Intelligence and Statistics*. 2019, pp. 2525–2534.

[5] B. Esmaeili, H. Huang, B. Wallace, and J.-W. van de Meent. "Structured Neural Topic Models for Reviews". In: *The 22nd International Conference on Artificial Intelligence and Statistics*. 2019, pp. 3429–3439.

[6] H. Chen, A. Kuzina, B. Esmaeili, and J. M. Tomczak. "Variational Stochastic Gradient Descent for Deep Neural Networks". In: *arXiv preprint arXiv:2404.06549* (2024).

Workshop

[1] B. Esmaeili, R. Walters, H. Zimmermann, and J.-W. van de Meent. "Understanding Optimization Challenges when Encoding to Geometric Structures". In: *NeurIPS 2022 Workshop on Symmetry and Geometry in Neural Representations*. 2022.

[2] H. Wu*, B. Esmaeili*, M. L. Wick, J.-B. Tristan, and J.-W. van de Meent. "Conjugate Energy-Based Models". In: *Energy Based Models Workshop - ICLR 2021*. 2021.

[3] H. Wu*, B. Esmaeili*, M. Wick, J.-B. Tristan, and J.-W. van de Meent. "Conjugate Energy-based Models". In: *Third Symposium on Advances in Approximate Bayesian Inference*. 2021.

[4] H. Zimmermann, H. Wu, B. Esmaeili, S. Stites, and J.-W. van de Meent. "Nested Variational Inference". In: *Third Symposium on Advances in Approximate Bayesian Inference*. 2021.

[5] A. Bozkurt, B. Esmaeili, D. H. Brooks, J. Dy, and J.-W. van de Meent. "Can VAEs Generate Novel Examples?" In: *NeurIPS Workshop on Critiquing and Correcting Trends in Machine Learning*. 2018.

EDUCATION

Eindhoven University of Technology *2023 – 2025*
Postdoc, Machine Learning
Advisor: Dr. Jakub M. Tomczak
Area: Deep Generative Models, Quantization

University of Amsterdam *2021 – 2023*
PhD, Machine Learning
Advisor: Dr. Jan-Willem van de Meent
Area: Deep Generative Models, Representation Learning

Northeastern University (*Transferred*) *2017 – 2021*
PhD, Computer Science
Advisor: Dr. Jan-Willem van de Meent
Area: Deep Generative Models, Representation Learning

University of Edinburgh *2016 – 2017*
MSc, Data Science
Grade: **Distinction** (above 70%)
Advisor: Prof. Michael Guttman
Dissertation: Bayesian Optimization for Likelihood Free Inference

University of Edinburgh *2012 – 2016*
BSc (Hons), Artificial Intelligence and Computer Science
Grade: **First Class** (above 70%)
Advisor: Prof. Michael Herrman
Dissertation: Particle Swarm Optimization

EXPERIENCE

Teaching Assistant
• DS-5230 – Unsupervised Machine Learning and Data Mining *Spring 2021*
• CS-7140 – Advanced Machine Learning *Spring 2018*

Research Assistant *Summer 2014*
University of Edinburgh
School of Informatics
Advisor: Prof. Paul Anderson
Project: Social media interaction models for teaching and learning

REVIEWING

Advances in Neural Information Processing Systems (NeurIPS) *2019 – 2023*
International Conference on Machine Learning (ICML) *2020 – 2023*
AAAI Conference on Artificial Intelligence (AAAI) *2020 – 2021*
International Conference on AI and Statistics (AISTATS) *2021 – 2023*
International Conference on Learning Representations (ICLR) *2021 – 2022*
International Journal of Approximate Reasoning (IJAR) *2022*

Awards:
International Conference on AI and Statistics (AISTATS) - Top 10% Reviewer Award *2023*
International Conference on Machine Learning (ICML) - Top 10% Reviewer Award *2021*
International Conference on Machine Learning (ICML) - Top 33% Reviewer Award *2020*

Advances in Neural Information Processing Systems (NeurIPS) - Top 50% Reviewer Award *2019*

RESEARCH INTERESTS

I am interested in deep generative models and how we can guide them towards learning useful representations for downstream tasks. Additionally, I am interested in the quantization of deep neural networks to optimize memory usage and improve inference efficiency. I am also an advocate of probabilistic programming, which I believe offers exciting possibilities for abstracting probabilistic models and general frameworks for inference.